Activity 2.6 – ATM Controller

1.Design Approach:

Designed an ATM controller using Verilog HDL with PIN verification, withdrawal processing, balance management and account locking functionality. Implemented logic to restrict access after repeated incorrect PIN attempts.

2.Verification Steps:

- Created a separate testbench.
- Applied reset to initialize the account balance.
- Tested correct and incorrect PIN entries.
- Verified withdrawal operations.
- Checked balance updates.
- Tested account locking functionality.
- Simulated using ModelSim.

3.Results:

The testbench checks PIN verification, withdrawal approval, balance updates and account locking. PASS/FAIL messages are displayed.

4.Issues Faced:

Incorrect PIN attempts must be counted accurately. Withdrawals must not exceed the available balance. Account locking conditions require proper implementation.

5.Conclusion:

Successfully designed and verified an ATM controller with PIN verification and withdrawal management using Verilog HDL.